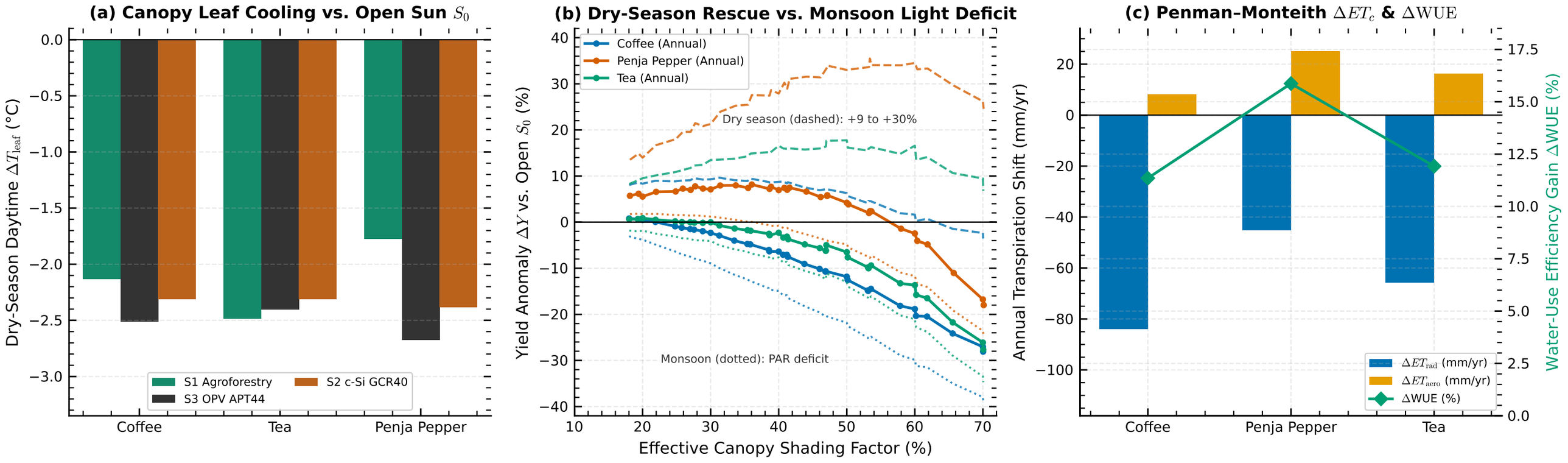

(a) Canopy Leaf Cooling vs. Open Sun S_0
Dry-Season Daytime ΔT_{leaf} (°C)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
S1 Agroforestry
S2 c-Si GCR40
S3 OPV APT44
Coffee
Tea
Penja Pepper
(b) Dry-Season Rescue vs. Monsoon Light Deficit
Yield Anomaly ΔY vs. Open S_0 (%)
40
30
20
10
0
−10
−20
−30
−40
Coffee (Annual)
Penja Pepper (Annual)
Tea (Annual)
Dry season (dashed): +9 to +30%
Monsoon (dotted): PAR deficit
10
20
30
40
50
60
70
Effective Canopy Shading Factor (%)
(c) Penman–Monteith ΔET_c & ΔWUE
Annual Transpiration Shift (mm/yr)
20
0
−20
−40
−60
−80
−100
Water-Use Efficiency Gain ΔWUE (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
ΔET_{rad} (mm/yr)
ΔET_{aero} (mm/yr)
ΔWUE (%)
Coffee
Penja Pepper
Tea